

National College of Ireland

Project Submission Sheet

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Programme: MSc In Cloud Computing

Year: 2026-2027

Module: Research In Computing CA1

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Submission Due Date:

12/06/2026

Project Title: Predictive Storage Cost Optimization Framework for Amazon S3 based on Intelligent Storage-Class Management

Word Count: 1602

Academic Integrity Certificate Verification Code: 670kpSElbi

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

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Signature: Varun Gampa

Date: 12/06/2026

PLEASE READ THE FOLLOWING INSTRUCTIONS:

1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
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AI Acknowledgement Supplement

Research in Computing

Your Name/Student Number	Course	Date
Varun Gampa	MSc in Cloud Computing	12/06/2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
ChatGPT	ChatGPT is used to look into the ideas in a detailed manner and also to check the feasibility	Chatgpt.com

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

[Insert Tool Name]
ChatGPT prompt : amazon S3 thesis ideas for deep dive knowledge on cost optimisation and lifecycle
ChatGPT Response : Generated research topic ideas related to Amazon S3 cost optimisation, storage lifecycle management, intelligent storage-class recommendation, and cloud cost forecasting, helping identify a feasible topic.

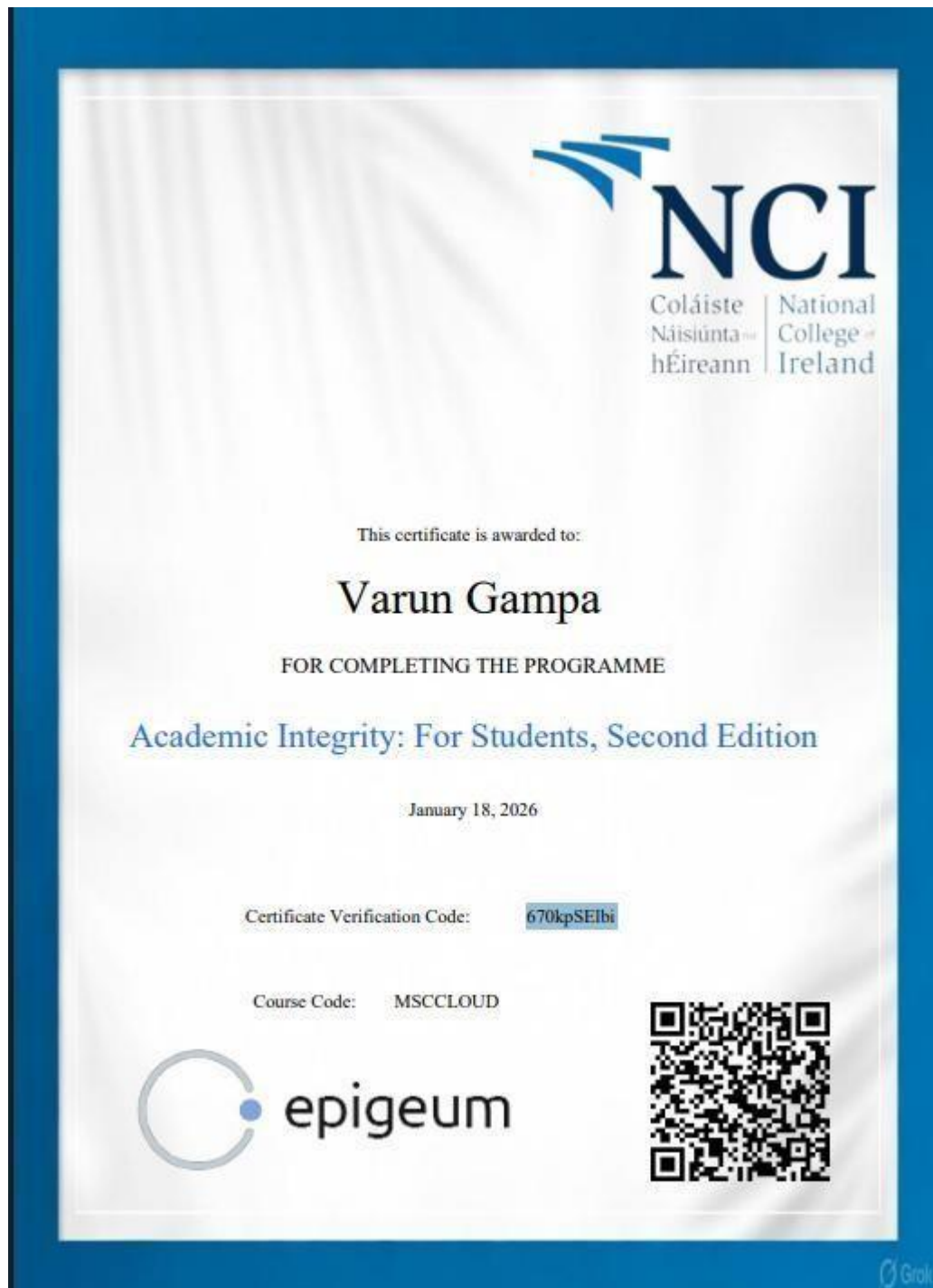
Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Place evidence here]

Academic Integrity Acknowledgement Supplement



Predictive Storage Cost Optimization Framework for Amazon S3 based on Intelligent Storage-Class Management

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Research in computing - CA1
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Research Problem Background

Organizations are relying on cloud computing as the main storage and management platform for huge and massive data sets. Amazon S3 is among the most widely used cloud storage object storage solutions that are available, with features such as scalability, durability, and compatibility with cloud-native applications. As more companies have increasingly embraced data intensive applications that run on the cloud, storage costs are a huge operating expense. This efficiency of data storage and making the data available has thus become a major challenge for cloud administrators and FinOps teams.

Amazon S3 has several storage classes: Standard, Standard-Infrequent Access (Standard-IA), Glacier Instant Retrieval, Glacier Flexible Retrieval, and Glacier Deep Archive. Each of these storage classes has a unique compromise of storage cost, retrieval speed and accessibility. The appropriate storage class for each object may result in cost savings. But, there are so many that are still storing data in expensive tiers for more than they need to, which results in unnecessary resource utilization and costs (Liu et al., 2025).

Amazon S3 Lifecycle Policies and Amazon S3 Intelligent-Tiering are AWS solutions to this problem. You can use Lifecycle Policies to move objects between storage classes based on a set of rules, such as object age, or Intelligent-Tiering automates the movement of objects between storage classes, based on access patterns that it detects. Most of these solutions are rule-based or reactive, and can help to save costs. Intelligent-Tiering only makes decisions after there has been a change in access pattern, while Lifecycle Policies do not take into account future access patterns.

Recent research in the use of intelligent approaches for placement and optimization of storage is discussed. This study has relied mainly on the research conducted by (Yang et al., 2025) on using machine learning technologies to enhance storage efficiency and decrease operational costs in warehouse-scale computing environments by making intelligent decisions in storage placement. Similarly, (Tu et al., 2025) investigated intelligent storage tiering mechanisms that aim to achieve high efficiency of storage hierarchy by making optimal decisions on data placement. Moreover, (Shen et al., 2025) suggested a new storage architecture called TierBase, which is cost-optimized and performance-oriented, by dynamically adjusting storage tier placement and optimizing data based on the workload.

Existing works prove the merits of intelligent storage placement, storage tiering, and workload-driven cost optimization, but do not particularly focus on providing Amazon S3 storage-class recommendation, storage cost forecasting and optimization savings estimation in one AWS-native framework, respectively (Yang et al., 2025; Liu et al., 2025; Tu et al., 2025; Shen et al., 2025). This is why a research gap exists in the field of developing a single solution for technical storage solutions optimization and financial forecasting for proactive cloud storage management through FinOps.

Therefore, an AWS-native framework is needed to support intelligent storage-class recommendation, storage cost optimization, and estimation of storage costs in the future. This research introduces a Predictive Storage Cost Optimization Framework for Amazon S3 that uses object characteristics and storage usage information to suggest appropriate storage classes, predict future storage costs and calculate the predicted savings. The proposed framework aims to be more efficient with the storage, establish FinOps practices and provide organizations with visibility of the cost of their cloud storage.

Research Question

What is the impact of implementing an integrated predictive storage cost optimization framework with intelligent storage-class recommendations, cost prediction and optimization savings estimation on storage utilization efficiency, cloud storage cost savings and proactive cloud FinOps-optimized decision making as compared to native AWS solutions?

Justification

The cloud storage management in a cloud environment like Amazon S3 is a big challenge in cloud computing because of the escalating cost of cloud storage management. With the massive amount of data being created and stored in big organizations, storage costs now make up a significant portion of the cost of operating a cloud. AWS offers storage optimisation features like Lifecycle Policies and Intelligent-Tiering, but these are rule based or reactive and don't offer much in terms of predicting future storage costs, or estimating the costs of optimisation choices (Yang et al., 2025). This makes it hard for organisations to make forward-thinking, cost-effective decisions about storage that optimise cost, performance and accessibility.

The proposed research is feasible and practical as it can be done by leveraging existing AWS technologies like Amazon S3, storage-class management features, AWS pricing information, Python, and the AWS SDK. The framework can be deployed and evaluated in a real AWS environment, allowing for a collection of operational data to be captured in a realistic environment, rather than just relying on simulated scenarios. This way, the results can be used directly in practical cloud storage setups, and successfully tested with experiments.

The effectiveness of the proposed framework can be measured by objective and quantifiable metrics like a reduction in storage cost, estimated cost savings, storage utilization efficiency, storage-class allocation accuracy, and effectiveness of storage optimization recommendations. These metrics offer an objective, quantitative means to assess the effectiveness of the framework and its ability to offer value over current AWS-native storage optimization options.

The research is valuable due to the rising importance of cloud cost optimization and FinOps in the minds of organizations looking to get the most out of their cloud investments. Recently, machine learning techniques have also been emphasized for enhancing the allocation and optimization of cloud resources, (Bodra, 2025). The proposed framework can aid in better understanding the storage spend, detect storage optimization opportunities, and predict future storage expenses, enabling more data-driven and informed cloud financial management practices. The results can contribute to organizations to optimize their budgeting, resource planning, cloud governance and ensure that they achieve adequate data accessibility and performance at data retrieval time.

The innovation of this research is the ability to integrate the three technologies – storage-class recommendation, storage cost estimation, and optimization saving estimation – within a single AWS-native framework. AWS storage optimization and storage cost reporting are solved by separate existing services like Lifecycle Policies, Intelligent-Tiering, and Cost Explorer. The proposed framework, however, combines these to provide one decision support system for managing cloud storage. This is an extension of existing knowledge, as it proposes a new way to find a solution to the synergy of intelligent storage optimization and financial forecasting for better technical and financial results in Cloud Computing environments.

The research also has an ethical aspect since it does not deal with sensitive personal data, human participants or high risk activities. All experimentation will be done with cloud storage resources and operational data created in a controlled AWS environment. Thus, the study poses very few ethical issues and provides benefits to organizations using cloud technologies.

The proposed framework will leverage Amazon S3 object metadata that is gathered via the AWS SDK (Boto3), such as object size, storage class, object age, and access frequency. The storage usage patterns will be analysed by a rule-based and machine learning-based recommendation engine, and suitable storage classes will be recommended. Storage usage and price history will be provided to create storage cost forecasts and estimate optimisation savings. The framework will be tested in a controlled AWS environment and measured by storage cost savings, storage utilisation efficiency, accuracy in storage-class allocation and estimated storage optimisation savings.

Specific Items to Be Addressed

Item #1: Create a metadata collection module with the AWS SDK (Boto3) that will gather the attributes of objects stored on Amazon S3 such as object size, storage class, object age, and access frequency, and store the data as a structured dataset for use with S3 storage analysis and cost optimization.

Item #2: Build an intelligent storage-class recommendation engine that uses metadata from S3 objects to make intelligent recommendations using a rules-based approach and data-driven methods to determine the most cost effective storage class for specific object characteristics, access patterns and storage needs.

Item #3: Develop a predictive storage cost forecasting feature based on historical storage usage and pricing information to estimate the cost of storage into the future and the potential optimisation savings available per storage class.

Item #4: Assess the effectiveness of the proposed framework by comparing storage-class allocation accuracy, storage utilisation efficiency, storage cost reduction, and estimated optimisation savings with the current storage configurations, and existing AWS-native solutions.

Item #5: Deploy and test the full solution in a controlled AWS environment and evaluate how effective the solution is in supporting cloud storage optimisation and proactive FinOps decision-making versus the current AWS native offerings.

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